CSE400 LEc 15-17 Joint Random Variables

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When you are writing the scribe, do reasoning step by step through definitions, assumptions, proofs, derivations, and worked example solutions exactly as presented in the provided context. Include this reasoning in the scribe so that the logical flow is clear for revision. Use only the provided lecture slides and relevant textbook excerpts. Don't mention any material, explanations or examples that are not present in the context.
Organize the scribe clearly in a formal academic style.

Below is a **formal lecture scribe for CSE400 Lecture 15–17: Joint Random Variables and Joint Distributions**, written strictly from the provided slides. The reasoning, definitions, derivations, and examples follow the logical flow of the lecture material for **exam preparation**.

Source:

CSE400 – Fundamentals of Probability in Computing

Lecture 15–17 Scribe

Joint Random Variables and Joint Distributions

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1. Motivation: Why Study Joint Random Variables?

In many real-world problems, we are interested in **two or more random variables simultaneously** rather than studying each variable independently. This allows us to analyze the **relationship or dependence between variables**.

Examples discussed in the lecture include:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and heart disease rate
- Air pollution level and respiratory illness rate
- Number of social media friends and age of users

These examples illustrate that **multiple variables may influence or relate to each other**, requiring joint probabilistic analysis.

Application Examples

Several computing and engineering applications were highlighted:

1. Smart Transportation

Let

- X = Vehicle Speed
- Y = Traffic Density

Key reasoning question:

Can vehicle speed be high when traffic density is also high?

This shows that **speed and density may be correlated**, which motivates the need for **joint random variables** in traffic analysis systems such as:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

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2. Wireless Network Performance

Let

- X = Signal-to-Noise Ratio (SNR)
- Y = Data Throughput

Question:

If SNR is high, will throughput always be high?

Reasoning:

Throughput depends on SNR but can also be affected by **network congestion**, indicating **conditional relationships between variables**.

Applications include:

- 5G scheduling
- Adaptive modulation
- Network planning

---

3. Other Applications

Additional examples presented in the lecture include:

- Weather prediction
  - X = Rainfall
  - Y = Temperature
- Drone delivery
  - X = Wind Speed
  - Y = Delivery Time
- Rolling two dice, where we analyze relationships such as $X + Y = 7$ or $X > Y$.

Key Concept

When multiple random variables are considered simultaneously:

- They possess a **joint distribution**
- This distribution allows computation of probabilities involving **both variables simultaneously**
- It helps analyze **dependence relationships**

The simplest situation occurs when variables are **independent**, but when they are not independent we measure relationships using:

- Covariance

- Correlation

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2. Discrete Case: Joint Probability Mass Function (Joint PMF)

Setup

Suppose we have two discrete random variables:

$$X = \{x_1, x_2, ..., x_n\}$$

$$Y = \{y_1, y_2, ..., y_m\}$$

The ordered pair (X, Y) takes values in the **product set**

$$\{(x_1, y_1), (x_1, y_2), ..., (x_n, y_m)\}$$

Thus, every possible outcome is a **pair consisting of values from both variables**.

Definition: Joint PMF

The **joint probability mass function** is defined as

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

Interpretation:

This represents the probability that **both events occur simultaneously**, i.e.,

- X takes value x_i
- Y takes value y_j

Joint PMF Table Representation

The joint PMF is often organized in a **two-dimensional table** where:

- Rows correspond to values of X
- Columns correspond to values of Y

Each cell contains the probability:

$$p(x_i, y_j)$$

Key Properties

1. Probabilities must lie between 0 and 1

$$0 \leq p(x_i, y_j) \leq 1$$

2. Total probability must equal 1

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

These conditions ensure that the joint PMF represents a **valid probability distribution**.

---

3. Example: Rolling Two Dice

Experiment

Roll two fair dice.

Define random variables:

$$X = \text{value on the first die}$$

$$T = \text{total sum of both dice}$$

Each ordered pair outcome (i, j) occurs with probability

$$\frac{1}{36}$$

because there are **36 equally likely outcomes** when rolling two dice.

Constructing the Joint Probability Table

The joint table lists probabilities for combinations of:

- X = first die value
- T = total sum

Observed properties from the table:

- Entries are either 0 or $1/36$

- Many combinations are impossible and therefore have probability 0

Example reasoning:

If $X = 1$, possible totals are:

$$T = 2, 3, 4, 5, 6, 7$$

These correspond to pairs:

$$(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)$$

Thus each valid pair contributes probability $1/36$.

Observation

From the table we see that:

$$X \text{ and } T \text{ are not independent}$$

because knowing the value of X restricts the possible values of T.

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4. Continuous Case: Joint Probability Density Function (Joint PDF)

The continuous case is analogous to the discrete case but with key differences:

Discrete Case	Continuous Case
Values are discrete	Values lie in intervals
Joint PMF	Joint PDF
Summation	Double integration

If

$$X \in [a, b], \quad Y \in [c, d]$$

then the pair (X, Y) lies in the **rectangular region**

$$[a, b] \times [c, d]$$

in the two-dimensional plane.

Definition: Joint PDF

A function $f(x, y)$ is called the **joint probability density function** if

$$P((X, Y) \text{ lies in a small rectangle}) \approx f(x, y)\, dx\, dy$$

Thus,

$$P((x, y) \in dx\, dy) = f(x, y)\, dx\, dy$$

Key Properties

1. Non-negativity

$$f(x, y) \geq 0$$

2. Total probability equals 1

$$\int_c^d \int_a^b f(x, y)\, dx\, dy = 1$$

Important note from the lecture:

The PDF **may take values greater than 1** because it is a **density**, not a probability.

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5. Worked Example

Given

$$f(x, y) = 4xy$$

for

$$0 \leq x \leq 1, \quad 0 \leq y \leq 1$$

Thus (X, Y) lies in the **unit square**

$$[0, 1]^2$$

Task 1: Show that $f(x, y)$ is a valid joint PDF

To verify validity we check **normalization**.

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

Step-by-step reasoning:

First integrate with respect to x:

$$\int_0^1 4xy \, dx = 2y$$

Now integrate with respect to y:

$$\int_0^1 2y \, dy = [y^2]_0^1 = 1$$

Since

$$\iint f(x, y) \, dx \, dy = 1$$

and $f(x, y) \geq 0$,

the function is a **valid joint PDF**.

Task 2: Compute $P(A)$

Event:

$$A = \{X < 0.5, Y > 0.5\}$$

Thus the probability is computed over the region:

$$0 \leq x \leq 0.5, \quad 0.5 \leq y \leq 1$$

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy \, dy \, dx$$

First integrate over y:

$$\int_{0.5}^1 4xy \, dy = 2x(1 - 0.25)$$

Then integrate over x:

$$\int_0^{0.5} 2x(1-0.25)dx$$
$$= \int_0^{0.5} \frac{3}{2}xdx$$
$$= \frac{3}{16}$$

Thus

$$P(A) = \frac{3}{16}$$

---

6. Joint Cumulative Distribution Function (Joint CDF)

Suppose X and Y are jointly distributed random variables.

The notation

$$X \leq x, \quad Y \leq y$$

represents the event

$$\{X \leq x \text{ and } Y \leq y\}$$

Definition

The **joint cumulative distribution function** is defined as

$$F(x, y) = P(X \leq x, Y \leq y)$$

This represents the probability that both variables lie **below specified thresholds simultaneously**.

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7. Joint CDF – Continuous Case

If X and Y are continuous with joint density $f(x, y)$, then the CDF is obtained by integrating the density:

$$F(x,y) = \int_c^y \int_a^x f(u,v)\,du\,dv$$

Recovering the Joint PDF

The density can be recovered by differentiation:

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

Thus:

- **CDF → integration**
- **PDF → differentiation**

---

8. Joint CDF – Discrete Case

If X and Y are discrete with joint PMF $p(x_i, y_j)$, then the CDF is obtained by **summing probabilities**:

$$F(x,y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j)$$

Interpretation:

To compute $F(x,y)$:

1. Add all probabilities in rows with $x_i \le x$
2. Add all probabilities in columns with $y_j \le y$

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9. Properties of the Joint CDF

The joint CDF $F(x,y)$ must satisfy the following properties:

1. Non-decreasing

If either x or y increases,

$$F(x,y)$$

must **stay constant or increase**.

2. Lower-Left Boundary

At the lower-left corner of the domain:

$$F(x, y) = 0$$

If the domain extends to $(-\infty, -\infty)$:

$$\lim_{(x,y)\to(-\infty,-\infty)} F(x, y) = 0$$

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3. Upper-Right Boundary

At the upper-right corner:

$$F(x, y) = 1$$

If the domain extends to (∞, ∞):

$$\lim_{(x,y)\to(\infty,\infty)} F(x, y) = 1$$

These properties ensure that the joint CDF behaves consistently as a **probability function**.

Key Exam Takeaways

Students should understand:

1. Motivation for **joint random variables**
2. Definition and properties of **joint PMF**
3. Table representation of joint distributions
4. **Joint PDF and normalization**
5. How to compute probabilities using **double integrals**
6. Definition and computation of **Joint CDF**
7. Relationship between **PDF and CDF**
8. Properties of the **joint CDF**

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professional lecture notes for exam revision.

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Note: The only lecture slides available in the provided context correspond to the material on **joint random variables, joint distributions, and related probability concepts**. The following scribe is therefore constructed strictly from those slides and their explanations.

CSE400 – Fundamentals of Probability in Computing

Lecture Scribe: Joint Probability and Conditional Relationships

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1. Introduction and Motivation

In many real-world problems, we must analyze **two or more random variables simultaneously** rather than studying each variable independently. This motivates the concept of **joint probability**, which describes the probability of events involving multiple variables occurring together.

According to the lecture slides, such situations arise frequently in science and engineering applications where multiple measurements are observed at the same time.

Examples discussed in the lecture include:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members

These examples demonstrate that **two variables may have a relationship**, meaning the value of one variable may provide information about the other.

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2. Real-World Motivating Applications

The lecture highlights several computing and engineering scenarios where joint probability modeling is required.

2.1 Smart Transportation

Two random variables are considered:

$$X = \text{Vehicle Speed}$$

$$Y = \text{Traffic Density}$$

Key reasoning question from the lecture:

Can vehicle speed be high when traffic density is also high?

The answer may vary depending on traffic conditions, which implies that **speed and density are correlated random variables**.

Applications include:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

2.2 Wireless Network Performance

In wireless communication systems:

$$X = \text{Signal-to-Noise Ratio (SNR)}$$

$$Y = \text{Data Throughput}$$

Throughput depends on SNR but may also be affected by other factors such as network congestion.

Thus, the relationship between these variables cannot always be determined independently, motivating the need for **joint probabilistic analysis**.

Applications include:

- 5G scheduling
- Adaptive modulation
- Network planning

2.3 Additional Examples

Other applications mentioned in the slides include:

- Weather prediction
  - X = Rainfall
  - Y = Temperature
- Drone delivery systems
  - X = Wind speed
  - Y = Delivery time
- Rolling two dice, where conditions such as $X + Y = 7$ or $X > Y$ may be analyzed using joint probability.

---

3. Joint Distribution Concept

When multiple random variables are studied together, they possess a **joint distribution**.

The joint distribution enables us to:

- Compute probabilities involving multiple variables

- Analyze relationships between variables
- Understand dependence structures

If variables are **independent**, the analysis becomes simpler.
When they are not independent, measures such as **covariance and correlation** are used to characterize their dependence.

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4. Discrete Case: Joint Probability Mass Function

4.1 Setup

Let two discrete random variables be defined as

$$X = \{x_1, x_2, \ldots, x_n\}$$

$$Y = \{y_1, y_2, \ldots, y_m\}$$

The pair (X, Y) takes values in the **product set**

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

Thus each outcome consists of a **pair of values**, one from each variable.

4.2 Definition: Joint PMF

The **joint probability mass function** is defined as

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

This quantity represents the probability that

- X takes the value x_i
- Y simultaneously takes the value y_j

Thus it measures the probability of the **joint occurrence of two events**.

---

5. Joint PMF Table Representation

The lecture presents a tabular representation for the joint PMF.

Structure of the table:

- Rows correspond to values of X
- Columns correspond to values of Y
- Each cell contains probability $p(x_i, y_j)$

Key Properties

1. Each probability must satisfy

$$0 \leq p(x_i, y_j) \leq 1$$

2. Total probability must equal 1

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

These properties ensure that the table represents a **valid joint probability distribution**.

---

6. Worked Example: Rolling Two Dice

6.1 Experiment

Roll two fair dice.

Define random variables:

$$X = \text{value on the first die}$$

$$T = \text{sum of the two dice}$$

There are 36 equally likely ordered outcomes.

Thus,

$$P((i, j)) = \frac{1}{36}$$

for each pair (i, j).

6.2 Constructing the Joint Table

A joint probability table is constructed for the variables X and T.

Observations from the table:

- Each entry is either 0 or $1/36$
- Certain combinations are impossible and therefore have probability 0

Example reasoning:

If

$$X = 1$$

then possible totals are

$$T = 2, 3, 4, 5, 6, 7$$

Each corresponds to outcomes

$$(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)$$

each with probability $1/36$.

6.3 Dependence Observation

From the constructed table:

$$X \text{ and } T \text{ are not independent}$$

because knowing the value of X restricts the possible values of T.

---

7. Continuous Case: Joint Probability Density Function

The continuous version of joint probability generalizes the discrete case.

The lecture states the following analogies:

Discrete Case	Continuous Case
Discrete values	Continuous intervals
Joint PMF	Joint PDF
Summations	Double integrals

If

$$X \in [a, b], \quad Y \in [c, d]$$

then the pair (X, Y) lies in the **rectangular region**

$$[a, b] \times [c, d]$$

in the two-dimensional plane.

Definition: Joint PDF

A function $f(x, y)$ is called the **joint probability density function** if

$$P((X, Y) \text{ lies in a small rectangle}) \approx f(x, y)\, dx\, dy$$

Thus the probability that the pair falls within a small region of area $dx\, dy$ is approximately

$$f(x, y)\, dx\, dy$$

Key Properties

1. Non-negativity

$$f(x, y) \geq 0$$

2. Total probability equals 1

$$\int_c^d \int_a^b f(x, y)\, dx\, dy = 1$$

An important note in the slides states that

- $f(x, y)$ may be greater than 1
- because it represents **probability density**, not probability.

---

8. Example: Continuous Joint PDF

Given

$$f(x, y) = 4xy$$

for

$$0 \le x \le 1, \quad 0 \le y \le 1$$

Thus the pair (X, Y) lies in the **unit square**

$$[0,1]^2$$

Step 1: Verify Validity (Normalization)

To check whether $f(x, y)$ is a valid PDF:

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

First integrate with respect to x:

$$\int_0^1 4xy \, dx = 2y$$

Then integrate with respect to y:

$$\int_0^1 2y \, dy = [y^2]_0^1 = 1$$

Thus:

- total probability = 1
- $f(x, y) \ge 0$

Therefore the function is a **valid joint PDF**.

Step 2: Compute Probability of an Event

Event:

$$A = \{X < 0.5, Y > 0.5\}$$

Thus the probability is computed over the region:

$$0 \le x \le 0.5, \quad 0.5 \le y \le 1$$

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy \, dy \, dx$$

Evaluating the integral yields

$$P(A) = \frac{3}{16}$$

---

9. Summary of Key Concepts

The lecture establishes the following fundamental ideas:

1. Joint probability models multiple random variables simultaneously.
2. The **joint PMF** describes discrete joint distributions.
3. The **joint PDF** describes continuous joint distributions.
4. Joint distributions allow computation of probabilities involving multiple variables.
5. Dependence between variables can be studied through joint distributions.

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CSE400 – Fundamentals of Probability in Computing

Lecture Scribe

Joint Probability and Conditional Relationships

---

1. Motivation: Why Study Joint Probability?

In many scientific and engineering problems, we are interested in analyzing **two or more random variables simultaneously** rather than studying each variable separately. This motivates the concept of **joint probability**, which allows us to compute probabilities of events involving multiple variables occurring together.

The slides emphasize that in real-life situations multiple variables often interact with one another.

Examples presented include:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and heart disease rate
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members

These examples highlight that the behavior of one variable may provide information about the other, motivating the need for **joint probability distributions**.

---

2. Applications of Joint Probability

2.1 Smart Transportation Systems

Two random variables are considered:

$$X = \text{Vehicle Speed}$$

$$Y = \text{Traffic Density}$$

A key question raised in the lecture:

> Can vehicle speed be high when traffic density is also high?

This question illustrates that **vehicle speed and traffic density are correlated random variables**.

Applications include:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

2.2 Wireless Network Performance

In wireless systems, two important variables are

$$X = \text{Signal-to-Noise Ratio (SNR)}$$

$$Y = \text{Data Throughput}$$

Throughput depends on SNR but may also be affected by network congestion and interference.

Thus:

- Even if SNR is high, throughput may not always be high.
- This indicates a **conditional relationship between variables**.

Applications include:

- 5G scheduling
- Adaptive modulation
- Network planning

2.3 Additional Real-World Examples

The lecture also discusses the following applications:

Weather prediction:

$$X = \text{Rainfall}, \quad Y = \text{Temperature}$$

Drone delivery systems:

$$X = \text{Wind Speed}, \quad Y = \text{Delivery Time}$$

Dice experiment:

Conditions such as

$$X + Y = 7$$

or

$$X > Y$$

require analyzing **joint probabilities** of the two variables.

---

3. Concept of Joint Distribution

When two random variables are studied together, they possess a **joint distribution**.

This distribution enables us to:

- Compute probabilities of events involving both variables
- Understand relationships between variables
- Analyze dependence structures

The lecture notes highlight that:

- If variables are **independent**, analysis is simpler.
- When they are **not independent**, we measure dependence using
  - covariance
  - correlation.

---

4. Discrete Case: Joint Probability Mass Function (Joint PMF)

4.1 Setup

Suppose X and Y are discrete random variables.

Let

$$X = \{x_1, x_2, \ldots, x_n\}$$

$$Y = \{y_1, y_2, \ldots, y_m\}$$

The ordered pair (X, Y) takes values in the **product set**

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Thus each outcome consists of a **pair of values**, one from each variable.

4.2 Definition of Joint PMF

The **joint probability mass function** is defined as

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

Interpretation:

This gives the probability that

- X takes value x_i
- Y simultaneously takes value y_j

Thus it represents the **probability of a joint outcome occurring simultaneously**.

---

5. Joint PMF Table Representation

The joint PMF can be organized into a **two-dimensional probability table**.

Structure:

- Rows correspond to values of X
- Columns correspond to values of Y
- Each cell contains probability $p(x_i, y_j)$

Key Properties

The table must satisfy two conditions:

Property 1

Probabilities must lie between 0 and 1:

$$0 \leq p(x_i, y_j) \leq 1$$

Property 2

Total probability must equal 1:

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

These conditions ensure that the table represents a **valid probability distribution**.

---

6. Worked Example: Rolling Two Dice

6.1 Experiment Description

Two fair dice are rolled.

Define random variables:

$$X = \text{value on the first die}$$

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There are **36 equally likely outcomes**.

Thus each ordered pair outcome (i, j) occurs with probability

$$\frac{1}{36}$$

6.2 Constructing the Joint Probability Table

A joint probability table is constructed for (X, T).

Observation from the table:

- Entries are either 0 or $1/36$
- Some combinations are impossible and therefore have probability 0

Example reasoning:

If

$$X = 1$$

then possible totals are

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These correspond to outcomes

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each with probability $1/36$.

6.3 Dependence Observation

From the constructed table we observe that

$$X \text{ and } T \text{ are not independent}$$

because knowing the value of X restricts the possible values of T.

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7. Continuous Case: Joint Probability Density Function

The continuous case generalizes the discrete case.

The lecture highlights the following correspondences:

Discrete Case	Continuous Case
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A function $f(x, y)$ is called the **joint probability density function** if

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Thus the probability that (X, Y) falls inside a small region of area $dxdy$ is approximately

$$f(x, y)dxdy$$

Key Properties

Property 1

$$f(x, y) \geq 0$$

Property 2

$$\int_c^d \int_a^b f(x, y)\, dx\, dy = 1$$

Important note from the slides:

The PDF value may exceed 1 because it represents **probability density rather than probability**.

---

8. Worked Continuous Example

Given

$$f(x, y) = 4xy$$

for

$$0 \le x \le 1, \quad 0 \le y \le 1$$

Thus (X, Y) lies in the **unit square**

$$[0,1]^2$$

Step 1: Check Validity (Normalization)

To verify that the function is a valid PDF:

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

First integrate with respect to x:

$$\int_0^1 4xy \, dx = 2y$$

Then integrate with respect to y:

$$\int_0^1 2y \, dy = [y^2]_0^1 = 1$$

Thus the total probability equals 1 and $f(x, y) \ge 0$.

Therefore the function is a **valid joint PDF**.

Step 2: Compute Probability of Event

Event:

$$A = \{X < 0.5, \ Y > 0.5\}$$

Thus the probability is computed over the region

$$0 \le x \le 0.5, \quad 0.5 \le y \le 1$$

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy \, dy \, dx$$

Evaluating the integral yields

$$P(A) = \frac{3}{16}$$

9. Key Takeaways

From this lecture, the following key ideas should be understood:

1. Joint probability models events involving multiple random variables.
2. The **joint PMF** describes discrete joint distributions.
3. The **joint PDF** describes continuous joint distributions.
4. Joint distributions allow analysis of relationships between variables.
5. Dependence between variables can be studied through joint distributions.

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